DISTRIBUTED PROCESSING OF SOUNDS ACROSS MOUSE AUDITORY CORTICAL REGIONS

by

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A THESIS

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In this optional section, you may choose to thank your committee, your friends, your family, and/or anyone who removed barriers for you and/or encouraged you to excel. You may also thank any sources of funding you received. Look at other students' theses to see examples.

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1. Introduction

The auditory cortex (AC) is a hierarchically and topographically organized region of the mammalian brain responsible for processing sound. Previous research has established that the primary auditory cortex subregion (AuP) is tonotopically organized, with neurons spatially arranged from low to high preferred frequencies across the cortical surface.[1] This organization has been extensively studied in mice and provides a reliable functional anchor for identifying auditory cortex during electrophysiological recordings.[1] Beyond AuP, the auditory cortex has several secondary subregions. This thesis focused on four: AuP, the dorsal auditory area (AuD), the ventral auditory area (AuV), and the posterior auditory area (AuPo). While AuP is well characterized, the functional roles of AuD, AuV, and AuPo remained comparatively understudied, particularly at the level of neural populations.[2]

How auditory information is distributed across cortical subregions remains an open question.[3] The literature reflects competing frameworks, including hierarchical, parallel, and distributed models, though these are not mutually exclusive.[4] Under a hierarchical model, information flows progressively from lower- to higher-order areas, with increasing representational complexity at each stage.[4] Under a parallel processing model, distinct subregions independently handle different sound features simultaneously.[4] Under a distributed

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[1] T. A. Hackett et al., "Linking Topography to Tonotopy in the Mouse Auditory Thalamocortical Circuit," Journal of Neuroscience 31, no. 8 (2011): 2983–95, https://doi.org/10.1523/JNEUROSCI.5333-10.2011.

[2] B. Joachimsthaler et al., "Quantitative Analysis of Neuronal Response Properties in Primary and Higher-Order Auditory Cortical Fields of Awake House Mice," European Journal of Neuroscience 39, no. 5 (2014): 904–18, https://doi.org/10.1111/ejn.12463.

[3] A. J. E. Kell et al., "A Task-Optimized Neural Network Replicates Human Auditory Behavior, Predicts Brain Responses, and Reveals a Cortical Processing Hierarchy," Neuron 98, no. 3 (2018): 630–44, https://doi.org/10.1016/j.neuron.2018.03.044.

[4] G. H. Recanzone and Y. E. Cohen, "Serial and Parallel Processing in the Primate Auditory Cortex Revisited," Behavioural Brain Research 206, no. 1 (2010): 1–7, https://doi.org/10.1016/j.bbr.2009.08.015.

model, representations are shared across regions without strong specialization.[5] Previous work in mice established that the auditory cortex has primary and higher-order fields with distinct thalamic inputs and response properties that parallel the core-belt organization described in other mammals, suggesting these frameworks apply to the mouse AC.[1] Distinguishing between them required comparing the representational structure, dimensionality, and predictive capacity of population activity across multiple subregions simultaneously, which required recording from all subregions within the same sessions.

Single-neuron studies of auditory cortex revealed much about frequency tuning and individual response properties, but failed to capture the collective structure of neural populations. Population-level analyses revealed representational geometry not visible at the single-unit level, including the shape of a neural manifold, its effective dimensionality, and the separability of stimulus representations.[6] The development of Neuropixels probes[7] made it feasible to record hundreds of neurons simultaneously across multiple cortical subregions, providing the data density these analyses required. Despite progress in understanding AuP, the dorsal, ventral, and posterior auditory subregions remained poorly characterized at the population level. It was not known whether these subregions encoded complementary or redundant information relative to AuP, nor how their representational geometry differed across stimulus types of varying acoustic complexity.[8] This thesis addressed these gaps by applying three complementary population

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[5] J. A. Winer and C. C. Lee, "The Distributed Auditory Cortex," Hearing Research 229, no. 1–2 (2007): 3–13, https://doi.org/10.1016/j.heares.2007.01.017.

[6] L. Feigin et al., "Sparse Coding in Temporal Association Cortex Improves Complex Sound Discriminability," Journal of Neuroscience 41, no. 33 (2021): 7048–64, https://doi.org/10.1523/JNEUROSCI.3167-20.2021.

[7] J. J. Jun et al., "Fully Integrated Silicon Probes for High-Density Recording of Neural Activity," Nature 551, no. 7679 (2017): 232–36, https://doi.org/10.1038/nature24636.

[8] T. A. Hackett, "Information Flow in the Auditory Cortical Network," Hearing Research 271, no. 1–2 (2011): 133–46, https://doi.org/10.1016/j.heares.2010.01.011.

analyses, encoding, decoding, and discriminability, to simultaneous multi-area recordings from mice passively presented with pure tones, amplitude-modulated noise, natural sounds, and speech syllables.

This thesis was guided by one primary research question: how do the auditory cortical subregions AuP, AuD, AuV, and AuPo differ in their ability to represent different sound types? This primary question was addressed through three sub-questions: (1) Do AC subregions differ in neural manifold structure and effective dimensionality across sound types? (2) How accurately can each subregion decode stimulus features from population activity? (3) How separable are neural population responses to different stimuli within each subregion?

2. Experimental Design and Data

2.1 Behavioral Paradigm

All data were collected from head-fixed mice in the Jaramillo Lab at the University of Oregon. Mice were head-fixed and positioned on a running wheel for the duration of each recording session; locomotion was not experimentally controlled, recorded, or aligned to neural data. Sounds were presented passively during electrophysiological recording: no behavioral task was required, and no rewards were contingent on the presented sounds. Stimuli were delivered in pseudorandom order, cycling through all sounds in a category before repeating, to minimize stimulus-specific neural adaptation.

2.2 Stimulus Design

Two independent datasets were collected from separate subject cohorts with different stimulus sets. The Natural Sounds Dataset included pure tones, amplitude-modulated white noise, and natural sounds. The Speech Dataset included synthetic speech syllables. Within the Natural Sounds Dataset, paradigms were presented in ascending order of acoustic complexity: pure tones first, then amplitude-modulated sounds, then natural sounds. This ordering served to confirm the presence of sound-responsive neurons via pure tone responses before proceeding to more complex stimuli.

2.2.1 Pure Tones

Pure tone stimuli consisted of 16 frequencies logarithmically spaced between 2 kHz and 40 kHz. Pure tones served as the simplest stimulus class and as an anchor for verifying tonotopic organization and confirming electrode placement within auditory cortex.

2.2.2 Amplitude-Modulated White Noise

Amplitude-modulated (AM) white noise stimuli consisted of 11 modulation rates logarithmically spaced between 4 Hz and 128 Hz. The modulation rate defines the frequency at which the amplitude of a white noise carrier fluctuates per second. AM stimuli represented an intermediate level of acoustic complexity, requiring temporal processing beyond simple frequency tuning.

2.2.3 Natural Sounds

Natural sound stimuli consisted of 20 recordings drawn from five environmental sound categories: Frogs, Crickets, Streamside, Bubbling, and Bees. Each category contained four distinct exemplars, resulting in 20 unique stimuli. All 20 were treated as independent in analyses. Category membership was retained as a grouping variable for visualization and within-versus-between category discriminability comparisons.

2.2.4 Speech Syllables

Synthetic speech syllables were generated along two acoustic dimensions: voice onset time (VOT) and formant transition (FT). VOT refers to the delay between the release of a stop consonant and the onset of vocal cord vibration, distinguishing voiced from unvoiced stops (e.g., /ba/ vs. /pa/). FT refers to the trajectory of the second formant frequency at syllable onset, distinguishing place of articulation (e.g., /ba/ vs. /da/). Stimuli were arranged along a grid defined by VOT and FT values of 0, 33, 67, and 100 (percent of maximum), yielding 12 unique syllables at the boundary and corner positions of this 2D acoustic space. The four canonical endpoint syllables correspond to /ba/, /da/, /pa/, and /ta/.

[Figure 1: Experimental Design figure svg file that as A: the experiment illustration, B: the histology slice and brain regions, C: the PT, AM, and natural sounds wavelength diagrams/sound diagrams (still needs to be made), and D: speech syllables diagram (needs to be fixed to the right tuple values (0, 33, 67, 100). Screenshot:

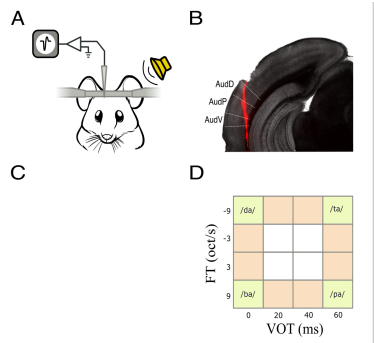

Figure 1: Experiment and Stimuli

Caption

2.3 Trials

For speech stimuli, the first 20 trials per unique FT/VOT token were retained, and trials were sorted into a consistent FT/VOT order across sessions and animals before concatenation. For non-speech stimuli, total trial counts were capped at load time: pure tones at 320 trials (20 per stimulus), AM at 220 trials (20 per stimulus), and natural sounds at 200 trials (10 per stimulus). When behavioral and electrophysiology trial counts differed by exactly one, the final electrophysiology trial was dropped; sessions with larger mismatches were excluded entirely.

2.4 Cortical Subregions

Neural data were recorded from four subregions of the auditory cortex, identified histologically based on electrode track location: AuP, AuD, AuV, and AuPo. AuP is the best-characterized subregion and served as the reference against which secondary subregions were

compared. AuD, AuV, and AuPo are thought to support higher-order auditory processing, though their specific functional roles remain unclear. AuD was recorded in Speech Dataset sessions but excluded from speech population figures and discriminability analyses due to insufficient neuron count for reliable subsampling; speech analyses therefore compared three subregions (AuP, AuV, AuPo).

3. Methods

3.1 Neural Recordings and Spike Sorting

Extracellular electrophysiological recordings were performed using Neuropixels 1.0 probes[8] inserted acutely into the auditory cortex of head-fixed mice. Each recording session involved a single probe penetration targeting multiple auditory subregions simultaneously. In some sessions the probe was repositioned within the craniotomy window to optimize subregion coverage, as cortical area boundaries span small distances and probe deflection during insertion can shift electrode positions relative to intended targets. Sound stimuli were time-locked to neural recordings using synchronized behavioral and electrophysiology systems.[9]

Raw extracellular voltage signals were processed using Kilosort for automated spike detection and clustering.[10] Spike clusters were then manually curated using Phy,[11] where each candidate unit was reviewed and classified as a well-isolated single unit or rejected based on waveform shape, inter-spike interval distributions, and cluster separation. Only well-isolated single units passing manual curation were retained, producing a set of spike timestamps for each accepted unit.

[Figure 2: PSTH and Raster Plots per sound with spike window boundaries marked. A: Pure Tones, B: AM, C: Natural Sounds, D: Combined Tuple Speech ALL raster plot and PSTH for

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[9] J. L. Mohn, M. M. Baese-Berk, and S. Jaramillo, "Selectivity to Acoustic Features of Human Speech in the Auditory Cortex of the Mouse," Hearing Research 441 (2023): 108920, https://doi.org/10.1016/j.heares.2023.108920.

[10] M. Pachitariu et al., "Fast and Accurate Spike Sorting of High-Channel Count Probes with KiloSort," Advances in Neural Information Processing Systems 29 (2016): 4455–63.

[11] C. Rossant et al., "Spike Sorting for Large, Dense Electrode Arrays," Nature Neuroscience 19, no. 4 (2016): 634–41, https://doi.org/10.1038/nn.4268.

one example cell with good variety of activity shown and clear spikes and not spikes (at least 100) — methods/plot_single_mouse_psth.py]

Figure 2: PSTH and Raster Plots with Spike Window Boundaries

Caption

3.2 Firing Rate Calculation

Spike timestamps were aligned to stimulus onset times from behavioral data files. For each neuron and each trial, spikes occurring within a defined post-stimulus time window were counted and divided by the window duration to yield firing rates in spikes per second. Trials were organized into a neuron-by-trial matrix for each session, subregion, and stimulus type, with each trial labeled by its corresponding stimulus identity.

Three non-overlapping time windows were defined per stimulus type to capture distinct phases of the neural response: onset, sustained, and offset. For pure tones, windows were onset [0–50 ms], sustained [50–100 ms], and offset [100–150 ms]. For AM sounds and speech syllables, windows were onset [0–200 ms], sustained [200–500 ms], and offset [500–700 ms]. For natural sounds, windows were onset [0–500 ms], sustained [1000–4000 ms], and offset [4000–4500 ms]. Analyzing all three windows per stimulus type preserved temporal response dynamics and allowed the analyses to capture how subregion encoding differed across distinct phases of the neural response.

3.3 Firing Rate Normalization

Raw HDF databases were converted into shared .npz firing-rate arrays by build_firing_rate_arrays.py, and those arrays were reused across all downstream analyses. Saved arrays included onset, sustained, and offset firing rates, along with brain-region labels, stimulus

labels, mouse IDs, and session IDs. For speech stimuli, firing rates were mean-centered per neuron across all trials within each spike window before saving the population arrays. This removed baseline differences in mean firing rate between neurons and sessions that would otherwise introduce noise into cross-session and cross-animal comparisons. For all analyses that required model fitting, predictor matrices were additionally z-scored using StandardScaler[12] before fitting.

3.4 Encoding

Encoding analyses characterized the structure of neural population representations across subregions and sound types. For all encoding analyses, neuron-by-trial matrices from all sessions and animals were concatenated along the neuron axis to form a single population matrix per subregion and stimulus type. This concatenation pooled neurons across the full dataset to maximize the resolution of population geometry estimates.

3.4.1 Principal Component Analysis

PCA was applied separately for each combination of sound type, brain region, and spike window. PCA was chosen because it provides an interpretable linear decomposition of population variance without requiring labeled data, making it well suited for exploratory characterization of representational geometry across subregions. Input matrices were mean-centered per neuron before PCA. Population PCA figures show all trials projected onto the first two principal components, with points colored by stimulus identity. Speech-specific figures were

[12] F. Pedregosa et al., "Scikit-Learn: Machine Learning in Python," Journal of Machine Learning Research 12 (2011): 2825–30.

additionally generated with points colored separately by FT value and VOT value to assess how each acoustic dimension was represented independently in population activity.

3.4.2 Participation Ratio

Effective dimensionality was quantified using the participation ratio, computed from the explained-variance spectrum of PCA.[13] Scree plots show the first 10 principal components (see Appendix B). Given a set of eigenvalues λ from PCA, the participation ratio is defined as $(\Sigma\lambda)^2 / \Sigma\lambda^2$. This scalar summarized how many principal components carried meaningful variance, with higher values indicating more distributed representations. The participation ratio is continuous and sensitive to the full shape of the variance spectrum, making it a good choice for a comparable metric across subregions and sound types. Because participation ratio scales with the number of neurons in the population, neurons were randomly subsampled to a fixed count before analysis (278 neurons for the natural sounds dataset, 99 for the speech dataset).

3.5 Decoding

3.5.1 Ridge Regression

To quantify how much information about stimulus identity was linearly decodable from population activity, ridge regression[14] was applied to predict acoustic stimulus features from neural responses at the session level. The high-dimensional, correlated structure of neural population data makes overfitting a central concern. L2 regularization shrinks all coefficients toward zero without zeroing any out, which preserves contributions from the full population and

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[13] S. Recanatesi et al., "A Scale-Dependent Measure of System Dimensionality," Patterns 3, no. 8 (2022): 100555, https://doi.org/10.1016/j.patter.2022.100555.

[14] A. E. Hoerl and R. W. Kennard, "Ridge Regression: Biased Estimation for Nonorthogonal Problems," Technometrics 12, no. 1 (1970): 55–67, https://doi.org/10.1080/00401706.1970.10488634.

is more appropriate than L1 regularization when decoding load is expected to be distributed across neurons rather than concentrated in a sparse subset.

Session-level rather than pooled-population decoding was used because it captured trial-to-trial variability within individual recordings and provided a natural unit of replication across animals and recording days. For pure tones and AM sounds, the regression target was the log-transformed stimulus parameter value, as these stimuli are logarithmically spaced.

Ridge regression decoding results are presented for pure tones and amplitude-modulated sounds. Natural sounds consisted of 20 exemplars drawn from five environmental categories (Frogs, Crickets, Streamside, Bubbling, and Bees) with no inherent acoustic ordering. Assigning a numeric regression target to exemplar identity would impose an arbitrary ranking that does not reflect any continuous or meaningful acoustic dimension, making a linear regression framework inappropriate for this stimulus type. Speech syllables are defined along two independent and orthogonal acoustic dimensions (VOT and FT). Ridge regression would collapse this 2D space onto an arbitrary ordering that does not preserve the perceptual or acoustic structure of the stimuli. For both natural sounds and speech, discriminability analysis (Section 3.5.2), which does not require an ordered regression target, was better suited to characterizing neural responses to these stimulus types.

The session matrices for each subregion and sound were z-scored independently and each session was subsampled 100 times to a fixed neuron count (10 neurons per session for speech, 30 neurons for non-speech sound types) to allow fair cross-region comparisons within each session. Each subsample was evaluated with outer shuffled 5-fold cross-validation, with alpha selected on training folds using RidgeCV[12] over a grid of 200 logarithmically spaced values from 10^{-10} to 10^{5}

(see Appendix C). The session-level score was the mean R^2 across subsamples. Figure x-axis labels report the number of balanced sessions (n), and panel text shows mean R^2, mean RMSE, and mean selected alpha (see Appendix D).

3.5.2 Discriminability

Discriminability analyses quantified how well population activity in each subregion separated pairs of stimuli. Like the encoding analyses, discriminability was computed on population matrices formed by concatenating neurons across all sessions and animals within each subregion and stimulus type. Predictor matrices were z-scored using StandardScaler[12] before analysis. Equal-neuron population counts were 278 neurons per region for the Natural Sounds Dataset and 99 neurons per region for the Speech Dataset, targets chosen to match the minimum available across regions.

Linear SVM discriminability used LinearSVC,[12] with hyper parameter C tuned separately for each combination of sound type, brain area, and spike window over a grid of 20 logarithmically spaced values from 10^{-5} to 10^{4} (see Appendix E). Pairwise classification accuracy was estimated using leave-one-out cross-validation[15]. Pairwise datasets were shuffled with deterministic seeds before classification. Linear SVM was chosen over nonlinear classifiers because the goal was to characterize linear separability of stimulus representations, which is directly relevant to whether downstream areas could read out stimulus identity with a simple linear decoder. Natural sound analyses additionally included a within-versus-between category comparison. Within-versus-between box plots include significance annotation bars colored in contrasting hues to visually distinguish within- from between-category comparisons.

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[15] M. L. Weese, B. J. Smucker, and D. J. Edwards, "The Use of Cross-Validation in the Analysis of Designed Experiments," arXiv preprint (2025), https://doi.org/10.48550/arXiv.2506.14593.

3.6 Statistical Analysis

Statistical comparisons were applied to summary distribution figures, focusing inference on the brain-region comparisons central to the thesis questions. Because recordings yielded unequal neuron counts across subregions, sessions, and animals, neurons were randomly subsetted to a fixed count before computing any summary statistic, ensuring that differences in distribution reflected differences in neural representation rather than differences in population size (see Appendix A).

For ridge regression box plots, all pairwise brain-region comparisons within each sound type and spike window were evaluated using Mann-Whitney U tests.[16] For all discriminability box plots, including region-comparison and natural sound within-versus-between category comparisons, pairwise comparisons were also evaluated using two-sided Mann-Whitney U tests, as these figures used unpaired comparisons across regions and stimulus pairs. Mann-Whitney U was chosen because it makes no assumption about the shape of the underlying distribution and is appropriate for the sample sizes typical of these comparisons. All p-values were corrected for multiple comparisons using Bonferroni correction.[17] Significance was annotated directly on figures using bracket-and-star notation, with stars shown only for corrected $p < 0.05$ (* for $p < 0.05$, ** for $p < 0.01$, *** for $p < 0.001$).

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[16] H. B. Mann and D. R. Whitney, "On a Test of Whether One of Two Random Variables Is Stochastically Larger than the Other," Annals of Mathematical Statistics 18, no. 1 (1947): 50–60, https://doi.org/10.1214/aoms/1177730491.

[17] O. J. Dunn, "Multiple Comparisons Among Means," Journal of the American Statistical Association 56, no. 293 (1961): 52–64, https://doi.org/10.1080/01621459.1961.10482090.

3.7 Code and Reproducibility

All analyses were implemented in Python and are publicly available at https://github.com/itszoetom/AudCompAnalysis. The repository is organized into analysis-specific folders for PCA, ridge, discriminability, and methods figures, with shared utilities in params.py, funcs.py, and plot_stats.py for parameters, cross-validation helpers, and statistical annotations respectively. Raw HDF databases were converted into shared .npz firing-rate arrays by build_firing_rate_arrays.py, and those arrays were reused across all analyses. Random subsampling and cross-validation procedures used fixed or deterministic seeds where applicable to support reproducibility across runs.[18]

4. Results

4.1 Encoding

{Figure 3: PCA 2D Projection Plots with PR labelled. A: Pure Tones, B: AM, C: Natural Sounds, D: combined tuple speech. 4 subregions for non speech, 3 for speech

Figure 3: Principal Component Analysis 2D Projections Across Cortical Regions and Sounds

Caption

PCA of population activity revealed that all four auditory cortical subregions formed structured neural manifolds across stimulus types, though the geometry of those manifolds differed by region and by sound category. For pure tones, each subregion followed a continuum-like distribution of trial projections along PC1, with points ordered by frequency from low to high across the principal axis. This tonotopic gradient was visible in all regions during both the onset and sustained windows. Participation ratios for pure tones were broadly similar across subregions, ranging from approximately 93 to 128, with AuP scree plots showing a comparatively flatter variance spectrum, indicating that its pure tone representations were distributed across a larger number of principal components than those in AuD, AuV, or AuPo.

For AM white noise, the population manifolds took on a curved, arc-like structure in AuP, AuV, and AuPo during the onset window, reflecting an ordered representation of modulation rate. This arc geometry largely collapsed in the sustained window, accompanied by a sharp drop in participation ratio for AuP and AuV relative to onset, suggesting that AM sustained representations are lower-dimensional and more structured. AuD showed no comparable arc geometry and little change in PR across windows, consistent with a more diffuse and less modulation-rate-ordered representation.

For natural sounds, all subregions showed categorical clustering in the PCA scatter plots, with exemplars from the same environmental category grouping together in the projection space. Participation ratios were uniformly lower for natural sounds than for either pure tones or AM noise across all subregions, consistent with the more discrete categorical structure of the stimuli. AuP and AuV both had sustained PRs below 40, reflecting highly structured and lower-dimensional representations. AuV showed the weakest clustering overall. AuPo showed visible cluster separation that was at least as strong as AuP across windows.

For speech syllables, PCA projections showed overlapping, low-structure distributions across all three subregions included in the speech dataset (AuP, AuV, AuPo). Participation ratios were similar across these regions, ranging from approximately 53 to 61, with only slight clustering visible in the onset window. No subregion produced clearly separated syllable representations in the low-dimensional projection.

4.2 Decoding

4.2.1 Ridge Regression

[Figure 4: Ridge Regression per Session R^2 box plots for Pure Tones and AM with MWU + Bonferroni statistical tests between each brain subregion combo. A: Pure Tones, B: AM]

Figure 4: Ridge Regression

Caption

Ridge regression decoding of log-transformed stimulus parameters from population activity showed subregion-specific patterns that differed across stimulus types. For pure tones, AuV produced the highest median per-session R^2 in the sustained and offset windows, with significant differences relative to AuD in the sustained window (Bonferroni-corrected Mann-

Whitney U test, $p < 0.001$) and relative to AuD and AuP in the offset window. AuD showed the lowest decoding performance across all windows, with median R^2 values near or below 0.05 in the sustained window compared to AuV medians near 0.30.

For AM white noise, AuP produced substantially higher median R^2 than all other subregions in the onset and sustained windows. Significant differences between AuP and AuD were observed in the onset window ($p < 0.01$) and between AuP and both AuD and AuV in the sustained window ($p < 0.001$). Mean R^2 across regions peaked in the sustained window at approximately 0.55 for AuP. AuD again showed the weakest performance, with median sustained R^2 near 0.40. AuPo and AuV occupied intermediate positions. No significant differences were observed in the offset window.

4.2.2 Discriminability

[Figure 5: Linear SVM Discriminability Classifier Accuracy. Some heat maps (like different ones or significant ones) and all the box plots. A: Pure Tones, B: AM, C: Natural Sounds, D: Speech. Natural Sounds also has the Within- vs. Between- plots. This could be broken into two figures with the heat maps and the box plots separated. Or simple sounds and complex sounds separated. Open to either. Statistical notations of MWU + Bonferroni]

Figure 5: Linear SVM Discriminability Classifier Accuracy Heat maps and Box Plots

Caption

Pairwise linear SVM discriminability of stimulus identity was high for both pure tones and AM white noise across all subregions, with most stimulus pairs classified well above chance. For pure tones, median pairwise accuracy was near ceiling across AuP, AuV, and AuPo in all windows, and all four regions showed strong off-diagonal structure in the heatmaps. A

significant difference was observed at onset between AuP and AuD ($p < 0.001$), with AuD showing lower pairwise accuracy for frequency-neighboring pure tone pairs. In the sustained and offset windows AuV showed the highest median accuracy, consistent with the ridge regression result. AuD showed relatively weaker discriminability for nearby frequencies across windows.

For AM white noise, pairwise discriminability was again strong across AuP, AuV, and AuPo. AuD was significantly weaker than the other three subregions at onset and offset ($p < 0.05$), and this weakness was most pronounced for pairs of similar modulation rates, visible as reduced accuracy near the diagonal of AuD heatmaps. This pattern was consistent with the lack of arc-like manifold structure observed in AuD encoding and with the lower AM ridge regression R^2 values.

For natural sounds, all subregions showed clear block-diagonal structure in the pairwise heatmaps, indicating that between-category pairs were easier to discriminate than within-category pairs. Across regions, between-category accuracy was significantly higher than within-category accuracy in all four subregions and all three windows ($p < 0.001$ in most comparisons). AuP showed the strongest within-category discriminability, with significantly higher within-category accuracy than AuD and AuV in the onset and offset windows ($p < 0.05$ and $p < 0.001$ respectively). AuPo showed a similar pattern to AuP in within-category discrimination. AuV showed the weakest natural sounds discriminability, with the lowest offset window accuracy and significant differences relative to AuP ($p < 0.001$) and AuPo in the offset window.

For speech syllables, discriminability was strongest in the onset window across all three included subregions (AuP, AuV, AuPo) and largely collapsed in the sustained and offset windows. AuP had the highest onset accuracy, significantly above AuV and AuPo ($p < 0.001$),

with heatmaps showing structured separability particularly for pairs of syllables differing in both VOT and FT. AuP strong onset performance was consistent with the lower-dimensional encoding observed for complex sounds in AuP and with the sustained PR drop visible in encoding. No significant subregion differences were observed in the sustained or offset windows.

5. Discussion

Across three complementary analyses applied to four auditory cortical subregions, the data support a stimulus-dependent distributed processing model in which subregions differ not in whether they respond to sound but in what aspects of sound they represent most effectively. No single subregion was uniformly dominant or uniformly weak across all stimulus types and analyses, but each subregion showed a consistent profile that distinguished it from the others.

AuP and AuPo were the strongest performers for complex sounds. AuP showed the highest AM ridge regression R^2, the strongest natural sounds within-category discriminability, and the highest speech onset discriminability. The low participation ratio of AuP during natural sound sustained responses, combined with high linear decodability, indicates that AuP forms compact, structured representations of complex acoustic categories. AuPo showed a similar profile for natural sounds, with strong decoding and clear categorical clustering in PCA, suggesting that AuP and AuPo may function as complementary areas for higher-order auditory representation. This is consistent with the core-belt organization described in other mammals, where areas receiving direct primary input tend to support finer-grained representations of complex stimuli.[6]

AuV showed a distinct and somewhat paradoxical pattern. Its pure tone decoding was the strongest of any subregion in the sustained and offset windows, and its AM onset arc geometry was well-formed, suggesting it is not simply a lower-fidelity copy of AuP. Yet for natural sounds and speech, AuV was the weakest region by multiple measures, showing the lowest sustained R^2, the lowest offset discriminability for natural sounds, and no advantage in speech. This combination suggests AuV may be specialized for simple, spectrally defined features rather than

for acoustic complexity. One possible interpretation is that AuV receives input weighted toward thalamic pathways that preserve spectral precision but not the temporal or categorical structure relevant to natural and speech sounds. This would be consistent with the parallel processing framework, in which subregions independently handle different feature dimensions rather than passing information through a strict hierarchy.[4]

AuD consistently showed the weakest encoding and discriminability for AM sounds, with no arc-like manifold structure in PCA and significantly lower pairwise accuracy for AM pairs at onset and offset. Participation ratios for AuD were numerically similar to other regions across stimulus types, arguing against a simple reduction in responsiveness. Instead, the AuD population appears to represent AM stimuli in a more distributed and less ordered format. This could reflect AuD's known connectivity with motor and frontal regions, which would be consistent with a role in integrating auditory input with behavioral context rather than in stimulus feature extraction.[2] AuD's relative strength for pure tone classification in the SVM onset window and its lower AM performance together suggest its responsiveness is not uniformly weak but is selectively attenuated for temporal modulation structure.

The speech results warrant some caution in interpretation. Overall R^2 values for ridge regression were near zero across all subregions, and discriminability was driven almost entirely by the onset window. This likely reflects the temporal structure of the synthetic syllables rather than a failure of the auditory cortex to represent speech: the stimuli were defined primarily by VOT and FT, features concentrated in the first 200 ms of the syllable, so sustained and offset windows would not be expected to carry much discriminating information. Within that constraint, AuP's onset discriminability advantage over AuV and AuPo suggests that the primary

area responds to the rapid acoustic transitions that define stop consonants with greater precision than secondary areas, consistent with its known sharpness of spectrotemporal tuning.

Taken together, these results do not cleanly resolve the debate between hierarchical and parallel processing models. Rather, they support a framework in which both types of organization are present simultaneously depending on the stimulus. For AM temporal modulation, a partial hierarchy is visible, with AuP outperforming AuD. For pure tone frequency encoding, a parallel arrangement is more appropriate, with AuV competitive with or superior to AuP. For natural and speech sounds, AuP and AuPo cooperate in a way that does not fit strictly hierarchical or strictly parallel descriptions. The between-subregion differences in representation structure, dimensionality, and linear decodability suggest that the four subregions divide representational labor in a stimulus-dependent way, with different regions contributing most at different levels of acoustic complexity.

Limitations of this work include small animal counts per subregion (n = 3 in most comparisons), which constrain the power of statistical comparisons and the generalizability of the ridge regression results in particular. The passive listening paradigm means it is not known whether these representational differences translate to differences in behavioral performance. Future work recording from the same animals under active discrimination tasks, or optogenetically silencing individual subregions, would help establish whether the functional specialization observed here is necessary for sound-guided behavior.

6. Conclusion

The auditory cortex does not process sound uniformly across its subregions. Using simultaneous Neuropixels recordings from AuP, AuD, AuV, and AuPo in passively listening mice, this thesis characterized how each subregion represents pure tones, amplitude-modulated white noise, natural sounds, and synthetic speech syllables at the population level. The results show that subregions differ in the geometry of their neural manifolds, the dimensionality of their representations, and the accuracy with which population activity can be decoded by a linear classifier. AuP and AuPo show the strongest representations of complex sounds, particularly for natural sound categorical structure and speech onset features. AuV is most effective for simple spectral features such as pure tone frequency. AuD shows selective weakness for AM temporal modulation encoding and discriminability across analyses. These patterns are consistent with a distributed processing model in which subregions divide representational labor depending on acoustic complexity, rather than simply relaying information through a single hierarchy or processing all features in parallel and independently. The population-level analyses applied here, encoding via PCA and participation ratio, linear decoding via ridge regression, and pairwise discriminability via linear SVM, each contributed a distinct and complementary view of subregion function that would not have been visible at the single-neuron level. This work establishes a population-level characterization of secondary auditory cortex in the mouse that can inform both computational models of auditory processing and the design of future experiments linking subregion activity to sound-guided behavior.

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8. Glossary

Bonferroni Correction: A multiple-comparisons correction that divides the significance threshold by the number of tests performed, controlling the family-wise error rate.

Firing Rate: The number of action potentials (spikes) a neuron produces per unit time, expressed in spikes per second (Hz), calculated by counting spikes within a defined window and dividing by window duration.

Leave-One-Out Cross-Validation: A cross-validation strategy in which one sample is held out as the test set and all remaining samples are used for training, repeated for every sample.

LinearSVC: A linear support vector classifier implemented in scikit-learn that finds the hyperplane maximizing the margin between classes. Used here for pairwise stimulus discriminability.

Mann-Whitney U Test: A nonparametric two-sample test that assesses whether one distribution tends to produce larger values than another, without assuming normality.

Participation Ratio: A scalar measure of effective neural population dimensionality computed from the explained-variance spectrum of PCA. Higher values indicate variance distributed across more principal components.

PCA (Principal Component Analysis): A linear dimensionality-reduction method that reprojects data onto orthogonal axes of greatest variance, enabling visualization and quantification of population structure.

Ridge Regression: A linear regression model with L2 regularization that penalizes large coefficients, reducing overfitting when predictors are high-dimensional or collinear.

Spike Window: A defined time interval relative to stimulus onset within which spikes are counted to compute firing rate. Three windows are used per stimulus type: onset, sustained, and offset.

5-Fold Cross-Validation: A cross-validation strategy that partitions data into five equal folds, training on four and testing on one, rotating until all folds have served as the test set.

StandardScaler / Z-scoring: A preprocessing step that subtracts the mean and divides by the standard deviation of each feature, producing zero-mean unit-variance inputs for model fitting.

9. Appendix

Appendix A: Data info plots

Neuron counts varied across brain regions, recording sessions, and animals within both datasets. The data info plots display these distributions as histograms of neurons per session, shown separately for each subregion and for the Natural Sounds Dataset and Speech Dataset. Because the analyses in this thesis relied on random subsampling to a fixed neuron count before any comparison, it was important to verify that all subregions contained sufficient neurons to support the target count across a reasonable number of sessions. These figures establish that the distribution of recorded neurons per session was unequal across regions, motivating the use of the fixed-count subsampling procedure described in Sections 3.4.2 and 3.5.2.

Histograms are shown separately per subregion, with the horizontal axis representing neuron count per session and bars reflecting session frequency at each count. The dashed vertical line in each panel marks the subsampling target (278 neurons for the Natural Sounds Dataset; 99 for the Speech Dataset), and sessions falling below this threshold were excluded from population analyses requiring that count.

[Figure A1: Neuron count histograms per subregion and session, Natural Sounds Dataset and Speech Dataset — methods/plot_data_info.py]

Figure A1. Distributions of neurons recorded per session across auditory cortical subregions for the Natural Sounds Dataset (AuP, AuD, AuV, AuPo; n = 6 mice) and the Speech Dataset (AuP, AuV, AuPo; n = 7 mice). Each bar reflects the number of sessions yielding a given neuron count within that subregion. Dashed vertical lines indicate the subsampling target used in

population analyses (278 and 99 neurons respectively). Sessions below the target were excluded from analyses requiring fixed-count populations.

Appendix B: Scree plots

Scree plots display the percent explained variance as a function of principal component (PC) index for the first 12 PCs, computed from the same population PCA runs described in Section 3.4.1. Each panel corresponds to one combination of subregion, spike window, and sound type, and the participation ratio (PR) is annotated within the panel. The scree plots complement the 2D PC scatter plots shown in the main PCA figure by making the full variance spectrum visible: a steeply falling scree indicates that variance is concentrated in a small number of dimensions (low PR), while a flat or gradually falling scree indicates more distributed structure (high PR). Scree plots were generated with the population PCA figure, and are compiled here for readability across conditions. They provide the quantitative basis for the PR comparisons reported in Section 4.1 and serve as a diagnostic for the quality of the low-dimensional projection shown in the main figure panels.

[Figure B1: Scree plots (percent explained variance vs. PC index, first 12 PCs) per subregion, spike window, and sound type, with participation ratio annotated — pca/plot_pca_population.py]

Figure B1. Scree plots showing percent explained variance as a function of PC index (PCs 1–12) for each auditory cortical subregion, spike window, and sound type. Bar height reflects the fraction of total population variance captured by each PC. The participation ratio (PR), defined as $(\Sigma\lambda)^2 / \Sigma\lambda^2$, is annotated in each panel. Higher PR values indicate more

distributed representations across principal components. Neuron counts were equalized across subregions before PCA (278 neurons, Natural Sounds Dataset; 99 neurons, Speech Dataset).

Appendix C: Ridge regression alpha hyperparameter tuning

For each session and subsample, the ridge regularization parameter α was selected on training folds using RidgeCV over a grid of 200 logarithmically spaced values from 10^{-10} to 10^{5}. The alpha tuning summary plots display the distribution of selected α values across sessions, subsamples, brain regions, and spike windows as boxplots, with one panel per spike window (onset, sustained, offset). These figures confirm that alpha selection was stable across conditions and that the regularization strength was not pathologically extreme in any subregion or window, which would suggest underfitting or overfitting in the ridge decoding results.

Each boxplot summarizes the distribution of selected α values across 100 subsamples and all balanced sessions for a given subregion and spike window. Values are plotted on a $\log_{10}$ scale. Consistency in selected α across subregions within a sound type and window indicates that the regularization was data-driven rather than dominated by edge effects of the grid.

[Figure C1: Boxplots of selected ridge α across sessions and subsamples, one panel per spike window (onset, sustained, offset), separately for pure tones and AM sounds — ridge/ plot_ridge_per_session.py]

Figure C1. Distribution of ridge regularization parameter α selected by RidgeCV across sessions and subsamples (100 per session), shown separately per subregion and spike window for pure tones (top) and AM white noise (bottom). α is plotted on a $\log_{10}$ scale. Each boxplot reflects the median and interquartile range of selected values across all balanced sessions and subsamples

for that subregion and window. Consistency in α across subregions indicates stable regularization behavior.

Appendix D: Ridge regression predicted vs. actual plots

The predicted vs. actual plots show scatter plots of ridge-predicted log-transformed stimulus parameter values against the true values, one panel per combination of brain region and spike window. A diagonal reference line indicates perfect prediction. These figures supplement the per-session R^2 boxplots in the main decoding results by showing the full distribution of individual trial predictions, making it possible to assess whether decoding errors are uniform across the stimulus range or concentrated at particular values (e.g., frequency extremes or specific modulation rates).

Each point represents a single trial prediction pooled across all sessions and subsamples for a given region–window combination. The horizontal axis is the true log-transformed stimulus parameter (log-frequency for pure tones; log-modulation rate for AM sounds) and the vertical axis is the model-predicted value. The diagonal reference line marks identity. Clustering of points near the diagonal indicates accurate decoding; systematic deviations indicate stimulus-specific prediction errors.

[Figure D1: Predicted vs. actual scatter plots per brain region and spike window for pure tones and AM sounds — ridge/plot_ridge_population.py]

Figure D1. Scatter plots of ridge-predicted vs. true log-transformed stimulus parameter values (log-frequency for pure tones; log-modulation rate for AM sounds) for each auditory cortical subregion and spike window. Each point is one trial prediction, pooled across sessions and subsamples. The diagonal line indicates perfect prediction. Panels are arranged by subregion

(rows) and spike window (columns). Tighter clustering around the diagonal indicates higher decoding accuracy.

Appendix E: Linear SVM C hyperparameter tuning

The SVM regularization parameter C was tuned separately for each combination of sound type, brain area, and spike window over a grid of 20 logarithmically spaced values from 10^{-5} to 10^{4}. The hyperparameter tuning plots show mean pairwise classification accuracy as a function of log C for each condition, allowing visual inspection of the tuning curve and confirmation that the selected C fell at or near the peak of each curve rather than at a grid boundary. Tuning was performed on the full balanced population before cross-validated classification.

Each panel plots mean pairwise accuracy on the y-axis against $\log_{10}$ C on the x-axis, with a vertical marker indicating the selected C value. A well-behaved tuning curve rises to a plateau and then remains flat or declines slightly at high C; a curve that peaks at the edge of the grid would suggest the search range should be extended. These plots confirm that the C grid was appropriate for all conditions tested.

[Figure E1: Mean pairwise SVM accuracy vs. log C for each combination of sound type, brain area, and spike window — discriminability/discriminability_analysis.py]

Figure E1. SVM hyperparameter tuning curves showing mean pairwise classification accuracy as a function of the regularization parameter C ($\log_{10}$ scale) for each combination of sound type, brain area, and spike window. The grid consisted of 20 logarithmically spaced values from 10^{-5} to 10^{4}. Vertical markers indicate the selected C value. Tuning was performed separately for each condition prior to final discriminability analyses.

Appendix F: Linear SVM example decision boundary plots

The decision boundary shows how the linear SVM separates pairs of stimuli in a reduced-dimensional population activity space. For selected stimulus pairs and brain regions, the first two principal components of the population response matrix are computed, and the linear decision boundary learned by LinearSVC is overlaid on the 2D scatter of trial projections. Each point represents a single trial, colored by stimulus identity, and the boundary line marks the hyperplane that maximally separates the two classes in that projection. These figures are intended as an illustrative complement to the pairwise accuracy heatmaps in the main discriminability results. They make visible the qualitative character of the separation: whether trial clouds are well-separated and compact, partially overlapping, or elongated in a particular direction relative to the boundary.

[Figure F1: Example linear SVM decision boundaries in 2D PC space for selected stimulus pairs and brain regions — discriminability/discriminability_analysis.py, discriminability/plot_linear_svm.py]

Figure F1. Illustrative linear decision boundaries in the first two principal component dimensions for selected pairwise stimulus comparisons and auditory cortical subregions. Each point represents one trial, colored by stimulus identity. The line indicates the linear decision boundary learned by LinearSVC on the full trial set for that pair. Panels were selected to represent a range of classification accuracy levels, spanning both simple (pure tone) and complex (natural sound) stimulus types.

Figures

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